MATH 3042- Lecture 7 (Review)

MATH 3042- Some review problems for Midterm Feb 2026

Please review all class notes and use this document as a supplementary study resource.

1. A network engineer monitors packet sizes (in KB) across 100 transmissions:

Min	Q1	Med	Mean	Q3	Max	Length	Sd
5.0	22.0	41.0	48.3	65.0	189.0	100	31.4

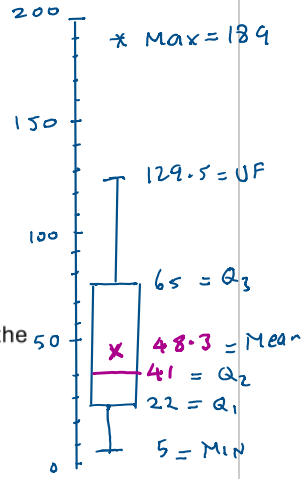
a. Calculate the upper and lower fences and identify any outliers.

b. Find the z-score of a packet size of 78 KB.

c. Is the distribution symmetric, right skewed, or left skewed?

d. A histogram column covering 160 ≤ Packet Size ≤ 180 has a frequency of 3. What percentage of transmissions fall there?

e. A packet size with z-scores above 2.5 will get flagged. What is the minimum size for the flagged packets?



a)

$IQR = Q_3 - Q_1 = 65 - 22 = 43$

$UF = Q_3 + 1.5(IQR) = 65 + 1.5 \times 43 = 129.5$ There are upper outliers.

$LF = Q_1 - 1.5(IQR) = 22 - 1.5 \times 43 = -42.5$ N/A There are no lower outliers.

b) $z = \dfrac{78 - 48.3}{31.4} = 0.95$

c) Right-skew: lower whisker shorter than the upper whisker.

Mean > Med

Upper outlier, but no lower outlier.

d) $\dfrac{f}{n} = \dfrac{3}{100} = 3\%$

e) Z-score $2.5 = \dfrac{x - 48.3}{31.4}$ $x = 126.8$ KB

Range of flagged:

$126.8 \leq x \leq 189$ KB

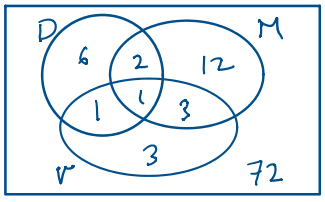
2. A machine learning training job can fail at the data loading stage (D), model training stage (M), or validation stage (V):

P(D) = 0.10, P(M) = 0.18, P(V) = 0.08, P(D ∩ M) = 0.03, P(D ∩ V) = 0.02, P(M ∩ V) = 0.04, P(D ∩ M ∩ V) = 0.01

Determine the probability that the training job **succeeds** (no stage fails).

$P(\bar{D} \cap \bar{M} \cap \bar{V}) = P(\overline{D \cup M \cup V})$

$= 100\% - 28\% = 72\%$



$$P(M|D) = \frac{P(M \cap D)}{P(D)} = \frac{0.03}{0.10} = 30\%$$
$$P(fail) = P(D \cup M \cup V) = 0.10 + 0.18 + 0.08 - 0.03 - 0.02 - 0.04 + 0.01 = 0.28 = 28\%$$

3. A containerized application can fail at the image build stage (I), the container orchestration stage (O), or the health check stage (H):

P(I) = 0.12, P(O) = 0.22, P(H) = 0.09, P(I | O) = 0.25, P(I ∩ H) = 0.02, P(O ∩ H) = 0.05, P(I ∩ O ∩ H) = 0.01

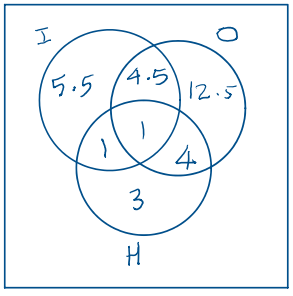
Determine the probability that the application **deploys successfully** (no stage fails).

$P(I \mid O) = \frac{P(I \cap O)}{P(O)}$, $0.25 = \frac{P(I \cap O)}{0.22}$, $P(I \cap O) = 0.055$

$P(I \cup O \cup H) = 31.5\%$

$P(\overline{I \cup O \cup H}) = 100\% - 31.5\%$

deploys successfully = $\boxed{68.5\%}$



with $n=20$: $1-\frac{P(300,20)}{300^{20}} = 100\% - 52.3\% = 47.7\%$

4. A distributed cache assigns 35 session tokens randomly into 300 hash buckets. What is the probability of at least one collision? How does this compare to the case with only 20 tokens in the same 300 buckets — which is more collision-prone and by how much?

35 tokens 300 buckets.

$P(\text{At least one collision}) = 1 - P(\text{no collision}) = 1 - \left(\frac{\text{we want} = P(300,35)}{\text{All} = 300^{35}}\right)$

$= 1 - 0.1269 \simeq 0.873 \simeq \boxed{87.3\%}$

Also same as: $1-\left(\frac{300}{300} \times \frac{299}{300} \times \frac{298}{300} \cdots \times \frac{266}{300}\right)$

with $n=35$ tokens the system is more collision-prone by about 40%.

5. A monitoring system checks 60 servers each hour. Each server has a 92% probability of responding without a timeout.

a. What is the probability that exactly 54 servers respond successfully?

b. What is the probability that at least 56 servers respond successfully?

c. Calculate the expected number of successful responses and the standard deviation.

d. Identify all unusually low numbers of successful responses using Z < −2.

Please use as needed:

dbinom(54, 60, 0.92) = 0.1454142

pbinom(54, 60, 0.92) = 0.3473999

pbinom(55, 60, 0.92) = 0.5298286

pbinom(56, 60, 0.92) = 0.7171439

Let x be the number of servers responding without timeout.

$0 \le x \le 60$ $x \sim Binom(p=0.92, n=60)$

a) $P(x=54) = dbinom(54, 60, 0.92) = 14.5\%$
b) $P(x \ge 56) = 1 - P(x \le 55) = 1 - pbinom(55, 60, 0.92) = 1 - 0.5298 \approx 47.02\%$
c) $\mu = E[x] = np = 60 \times 0.92 = 55.2$ $var = npq = 60 \times 0.92 \times 0.08 = 4.416$ $\sigma = \sqrt{4.416} = 2.1\ servers$
d) $-2 = \frac{x-55.2}{2.1}$, $x \approx 50.997$

If number of "success" is less than 51, we are having an unusual day

6. A Kubernetes cluster runs 100 scheduled jobs nightly. Each job has a 96% chance of completing successfully.

a. What is the probability that exactly 93 jobs complete successfully?

b. What is the probability that at least 97 jobs complete successfully?

c. Calculate the expected number of successful jobs and the standard deviation.

d. Identify all unusually low numbers of successful jobs using $Z < -2$.

Please use as needed:

```
> dbinom(93, 100, 0.96)
[1] 0.05888027
> pbinom(93, 100, 0.96)
[1] 0.1063923
> pbinom(96, 100, 0.96)
[1] 0.5705244
> pbinom(95, 100, 0.96)
[1] 0.3711359
```

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7. A software repository has 30 packages, 9 of which have known dependency conflicts. A developer randomly selects 6 packages to install. What is the probability that at least 1 has a dependency conflict?

9 have dependency conflict	Zero with conflict	
21 are free of conflict	6 free of conflict	

30 6

Let x be the number of packages from sample that have dependency conflict

$0 \leq x \leq 6$ $x \sim$ hypergeom

$P(x \geq 1) = 1 - P(x = 0)$

$= 1 - \dfrac{C(9,0) \times C(21,6)}{C(30,6)}$

$= 1 - \dfrac{54264}{593775} = 1 - 0.091 = 0.909$

$\simeq \boxed{90.9\%}$

bad 17
good 33
Total 50

10

8. A firewall ruleset contains 50 rules, 17 of which are misconfigured. A security auditor randomly samples 10 rules. What is the probability that at least 2 are misconfigured? Write down each step of what you are calculating.

let x be the number of misconfigured rules in the sample $0 \leq x \leq 10$

$$P(x \geq 2) = 1 - \left(P(x=0) + P(x=1) \right)$$

$$= 1 - \left(\frac{C(17,0)C(33,10)}{C(50,10)} + \frac{C(17,1) \times C(33,9)}{C(50,10)} \right) = 1 - 0.0728 = 0.927 \simeq 92.7\%$$

9. For each scenario, identify the most appropriate distribution and explain why:

a. A load balancer forwards requests to servers. On average, a server becomes overloaded 3 times per hour. What is the probability of exactly 5 overload events in the next hour? Poisson

b. A software team has 12 developers, 5 of whom are proficient in Rust. If 4 developers are randomly assigned to a new project, what is the probability that exactly 2 are Rust-proficient? hypergeom

c. A containerized microservice has a 20% chance of failing its health check on any given startup attempt. What is the probability that the first failure occurs on exactly the 3rd attempt? Geometric

d. A cloud function is invoked 200 times per day, each with a 99% success rate. What is the probability that exactly 195 invocations succeed? Binomial

e. A DevOps engineer deploys code 5 times per day on average. What is the probability of more than 7 deployments in a single day? Poisson

f. A test suite has 300 tests, 45 of which are currently failing. A reviewer randomly picks 15 tests to inspect. What is the probability that exactly 4 are failing? Hypergeom

g. A database connection attempt has an 8% chance of timing out. What is the probability that the first timeout happens on the 6th attempt? Geometric

h. An API gateway processes 500 requests independently, each with a 97% success rate. What is the probability that at least 490 succeed? Binomial

x: wait time between two consecutive crashes. $0 \le x < \infty$ $x \sim exp(Beta = 15)$

10. The time (in minutes) between crashes of an unstable microservice is exponentially distributed with a mean of 15 minutes.

a. What is the probability that the service runs for less than 10 minutes without crashing?

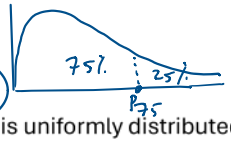
$P(x \le 10) = 1 - e^{-\frac{10}{15}} = 48.7\%$

b. What is the probability that the service runs between 12 and 20 minutes without crashing?

$P(12 \le x \le 20) = (1 - e^{-\frac{20}{15}}) - (1 - e^{-\frac{12}{15}}) = 0.1857 \simeq 18.6\%$

c. What is the 75th percentile of crash-free run times?

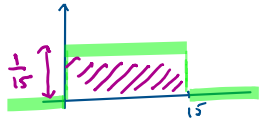
$1 - e^{-t/15} = 0.75$, $0.25 = e^{-t/15}$, $Ln(0.25) = -\frac{t}{15}$, $t \simeq 20.8$ minutes



11. The time (in seconds) a containerized function takes to ~~cold-start~~ is uniformly distributed between 0 to 15 seconds. $0 \le x \le 15$ $x \sim Unif(0, 15)$

a. What is the probability that a cold start completes in under 2 seconds?

$P(x < 2) = \frac{2}{15} \simeq 13.3\%$

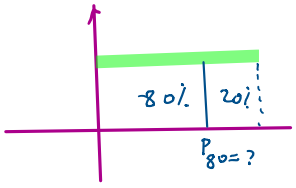


b. What is the probability that a cold start takes between 3 and 6 seconds?

$P(3 < x < 6) = \frac{6-3}{15} = \frac{3}{15} = 20\%$

c. What is the 80th percentile of cold-start times?

$15 \times 0.8 = 12$ seconds



$E[X] = \mu = \frac{b+a}{2}$

$Var = \frac{(b-a)^2}{12}$

12. The number of API calls per minute made to a microservice is normally distributed with a mean of 320 calls and a standard deviation of 45 calls.

a. What is the probability that in a randomly chosen minute, more than 400 calls are made? Place relevant numbers on a normal curve and shade the region. Write the filled-in R command.

b. What values mark the middle 90% of per-minute call volumes?

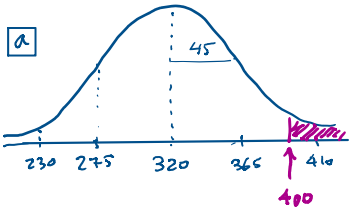
Given: qnorm(0.05) = -1.644854, qnorm(0.10) = -1.281552

X = number of calls in a random minute

$X \sim Norm(\mu = 320, \sigma = 45)$

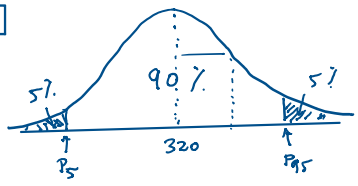
$P(x > 400) = 1 - pnorm(400, 320, 45)$

a



b) $\pm 1.644854 = \frac{x - 320}{45}$ → $x_{high} = 394$ calls, → $x_{low} = 246$ calls

b



13. The size (in MB) of log files generated per hour by a web server is normally distributed with a mean of 850 MB and a standard deviation of 120 MB.

a. What is the probability that a randomly chosen hour produces between 700 MB and 1000 MB of logs? Place relevant numbers on a normal curve and shade the region. Write the filled-in R command.

b. What values mark the middle 95% of hourly log sizes?

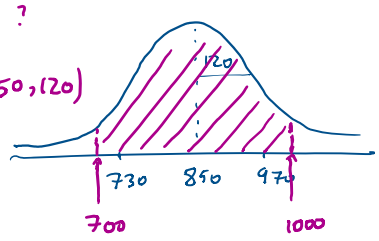
c. Storage alerts trigger when log size falls in the top 5%. What is the minimum log size that triggers an alert?

Given: qnorm(0.025) = -1.959964, qnorm(0.05) = -1.644854

0.89435 ?

$P(700 < x < 1000) = \text{pnorm}(1000, 850, 120) - \text{pnorm}(700, 850, 120)$

$= 0.89435 - 0.10565 = 0.7887$



14. Historically, 55% of users of a mobile app enable push notifications. A researcher surveys 180 users.

a. Calculate the mean and standard deviation of the sampling distribution of proportions.

b. Using the normal approximation, calculate the probability that more than 60% of the sampled users have push notifications enabled. Write the filled-in R command.

c. At a 5% significance level, is it unusual to observe only 45% enabling push notifications in this sample? Use qnorm(0.05) = -1.644854.

$P = 0.55$

$n = 180$

$np \geq 10 \qquad 180(0.55) \geq 10$

$nq \geq 10 \qquad 180(0.45) \geq 10$

CLT:

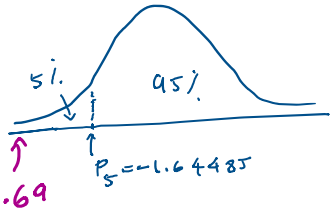
$\hat{p}$ is distributed normally

$\mu_{\hat{p}} = P = 0.55$

$\sigma_{\hat{p}} = SE = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.55 \times 0.45}{180}} = 0.037$

$z_{\hat{p}=0.45} = \frac{0.45 - 0.55}{0.037} = -2.69$

b) $P_r(\hat{P} > 0.60) = 1 - \text{pnorm}(0.60, 0.55, 0.037)$



$z = -2.69$

very unusual

15. Historically, 28% of cloud jobs exceed their allocated memory limit. A platform engineer samples 250 jobs.

a. Calculate the mean and standard deviation of the sampling distribution of proportions.

b. Using the normal approximation, what is the probability that between 25% and 33% of sampled jobs exceed their memory limit? Write the filled-in R commands.

c. At a 5% significance level, is it unusual to observe 35% of jobs exceeding their memory limit? Use qnorm(0.95) = 1.644854 in your reasoning.

Let $\hat{p}$ be the percentage of 250 cloud jobs that exceed allocated memory limit.
Check: $250 \times 0.28 \geq 10\ yes, \quad 250 \times 0.78 \geq 10\ yes$
CLT applies:
$\hat{p}$ ~ normal($\mu_{\hat{p}} = 0.28$, $\sigma_{\hat{p}} = SE = \sqrt{\frac{0.28 \times 0.72}{250}} = 0.02839718$)

b. $P(0.25 < \hat{p} < 0.33) = pnorm(0.33, 0.28, 0.028397) - pnorm(0.25, 0.28, 0.028397)$
c. z-score: $\frac{0.35-0.28}{0.028397} = 2.465033$

This is to the right side of the 95th percentile. It is unusually high.